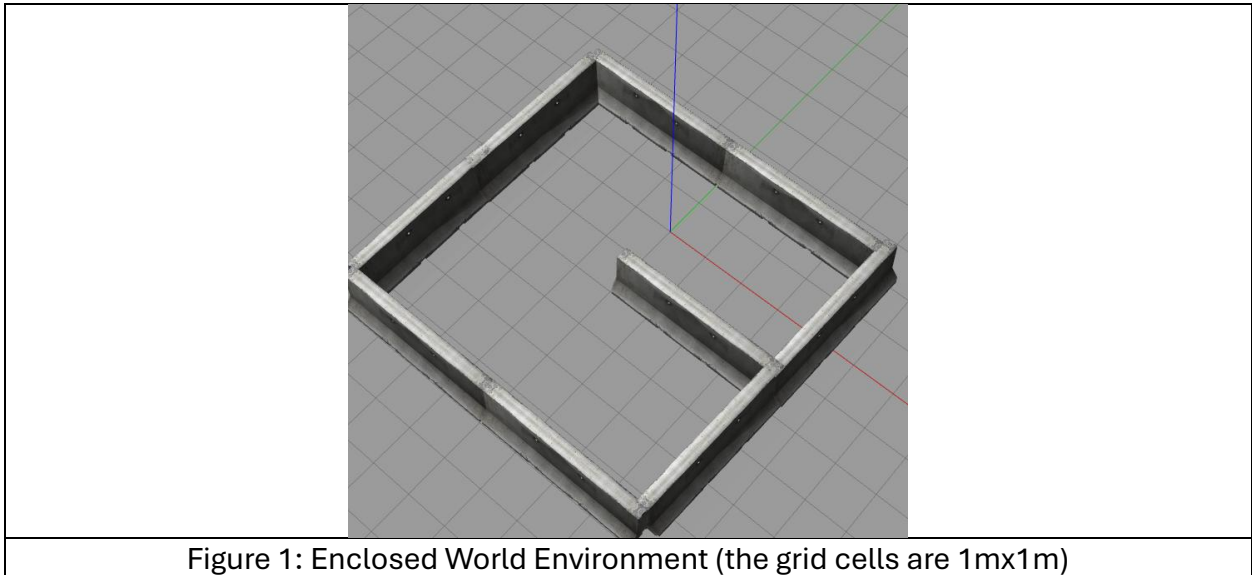


CISC 6525

Class project

Your class project is to build a probabilistic reasoning system to estimate **traversability** for a robot moving around in an enclosed space where the ground terrain varies in slipperiness. Traversability is high when slipperiness is low. The terrain on which the robot will move is flat and smooth, however, in some areas the wheels slip as the robot tries to move. The enclosed world environment consists of a 6 meter by 8 meter (approx.) space enclosed by Jersey barriers (Fig.1). This wheel slip can be measured by the robot. Your job is to generate a 2D map of the terrain that shows the probability that the Boolean random variable Traversable is True for each location. You can discretize the locations into a 0.5m x 0.5m grid covering the enclosed area.



You **must** use the following approach: Traversability_t is a hidden, Boolean random variable that must be estimated at time t . The robot location (x,y) at time t can be measured. You can assume that the prior for Traversability is $\langle 0.5, 0.5 \rangle$. You can also assume that there are contiguous regions of like traversability.

Wheel slip can be measured as a continuous random variable, Wheelslip_t . This is influenced by the Traversability_t random variable. You should propose a Dynamic Bayesian network structure for this problem, including the CPTs or other method to represent the probabilistic relationship of any node to its parents. (You are not constrained to use only these two random variables.)

You should propose and implement a state estimation algorithm (filtering) to estimate traversability and populate a 2D array with the probability of the traversability of every spot that the robot can reach (e.g., Fig. 2).

0.08	0.08	0.08	0.08	0.08	0.08
0.08	0.7	0.7	0.7	0.002	0.08
0.08	0.7	0.08	0.08	0.7	0.08
0.08	0.7	0.08	0.08	0.7	0.08
0.007	0.08	0.007	0.007	0.08	0.007
0.007	0.007	0.007	0.007	0.007	0.007

Figure 2: Example Traversability Table

You should implement a robot motion algorithm to move the robot throughout the space and generate the probability map. You could base this on the goto.py program and path-planning example covered earlier in class – of course you do not need to plan a path, you could hand code a path that drives the robot over the entire enclosed area.

Is it sufficient to cover each spot once to generate a good estimate of traversability?

You need to submit

1. A report showing and explaining your DBN and its design, state estimation and map generation design (don't include code here), your experimentation and your results.
2. Screenshots showing your program running and printing traversability
3. The final output matrix
4. All your code.

Rubric:

1. Is your DBN correct?
2. Do you implement state estimation correctly correctly?
3. Does your report include all the components mentioned and are your explanations well written and understandable?
4. Do your screenshots support a correctly working implementation?
5. Is your final output matrix in the form requested?
6. Does your final output support a correctly working implementation?
7. Does your code execute correctly in the container?